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Detection & Identification of Novel Attacks in Phishing using AI Algorithms

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Abstract

Ever since the inception of the internet, cyber security threats have been a big concern for the world. In 2022, more than 300,000 people fell victim to cyber-attacks accumulating a total loss of $52 million in the United States alone. As a result, the field of cyber security has burgeoned into a thriving $200 billion industry. Within this realm of cyber security threats, the most prevalent remains phishing. Governments, corporations and individuals alike have built safeguards in their systems to keep phishing threats at bay. Currently, anti-phishing techniques that have been predominantly researched and used in software products include list-based and visual similarity-based approaches. While these techniques provide excellent accuracy against previously known phishing attacks, they offer subpar accuracy against zero-day or novel attacks. Acknowledging this gap, our project, "Detection and Identification of Novel Attacks in Phishing using Artificial Intelligence Algorithms," endeavors to elevate the detection precision for novel/zero-day phishing incidents. Our approach includes the use of visual similarity methods for enhanced detection where it analyzes website screenshots. Our research started from a state-of-the-art dataset which we further enhanced marking a significant stride. We have meticulously augmented our dataset by sourcing and analyzing reported phishing pages of globally recognized brands from PhishTank, subsequently pairing these with screenshots of the corresponding legitimate websites. This strategic enrichment process aims to capture the sophisticated visual discrepancies between legitimate and phishing webpages, thereby facilitating the training of our model with a more comprehensive and representative dataset. Coupled with dataset enhancement, we have explored a diverse array of algorithms to ascertain the most effective approach for novel phishing detection. Through these concerted efforts, we aspire to not only advance the precision of phishing detection algorithms but also contribute a robust solution that addresses the evolving landscape of cyber security threats.

Executive Summary

Phishing continues to be a pervasive threat in the digital age, significantly eroding trust in online interactions. Despite advancements in security measures to counteract known threats, the emergence of zero-day attacks, exploiting undiscovered vulnerabilities, necessitates innovative detection strategies. This paper introduces a novel methodology centered on visual similarity to detect previously unrecognized phishing attempts effectively.

A pivotal advancement in our research is the strategic augmentation of our dataset. Instead of solely relying on screenshots from Pakistani-origin websites, which presented limitations in phishing page availability, we have adopted a more dynamic approach. By leveraging PhishTank, we meticulously curated a dataset comprising reported phishing pages targeting renowned global brands alongside corresponding legitimate website screenshots. This enhancement not only diversifies our dataset but significantly improves its representation of real-world phishing scenarios, offering a more robust foundation for our model's training process.

Our proposed model, utilizing three input tensors—an anchor, a positive, and a negative image—employs a two-stage methodology. Initially, it classifies websites as phishing or legitimate, followed by an optimization phase fine-tuning the model with misclassified examples. This dual-phase approach, coupled with our dataset's enrichment, markedly increases the model's accuracy in identifying novel phishing threats.

This comprehensive strategy underscores our commitment to advancing phishing detection capabilities. By enriching the dataset and refining our visual similarity-based model, we aim to significantly enhance the adaptability and effectiveness of phishing detection mechanisms in safeguarding against the evolving landscape of cyber security threats.

Table of Contents

List of Figures viii

List of Tables ix

1 Introduction 1

1.1 Purpose of this Document . . . 1

1.2 Intended Audience . . . 2

1.3 Definitions, Acronyms, and Abbreviations . . . 2

1.4 At the End of Introduction Chapter . . . 2

2 Project Vision 3

2.1 Problem Domain Overview . . . 3

2.2 Problem Statement . . . 4

2.3 Problem Elaboration . . . 4

2.4 Goals and Objectives . . . 4

2.5 Project Scope . . . 5

2.6 Sustainable Development Goal (SDG) . . . 5

2.6.1 Industry, Innovation and Infrastructure . . . 5

2.6.2 Peace, Justice and Strong Institutions . . . 6

3 Literature Review / Related Work 7

3.1 Definitions, Acronyms, and Abbreviations . . . 7

3.2 Detailed Literature Review . . . 7

3.2.1 VisualPhishNet: Zero-Day Phishing Website Detection by Visual Similarity . . . 7

3.2.2 An Effective Phishing Detection Model Based on Character Level Convolutional Neural Network from URL . . . 9

3.2.3 Intelligent Phishing Detection Scheme Using Deep Learning Algorithms . . . 10

3.2.4 Phishing Detection Using Machine Learning Techniques . . . 11

3.2.5 A Hybrid Phishing Detection System Using Deep Learning-based URL and Content Analysis . . . 12
3.2.6 High Accuracy Phishing Detection Based on Convolutional Neural Networks . . 13
3.2.7 An Intelligent Cyber Security Phishing Detection System using Deep Learning Techniques . . . 13
3.2.8 A Novel Phishing Detection Model Based on Revised Multi objective Evolution Optimization Algorithm and Random Forest . . . 14
3.2.9 An In-Depth Benchmarking and Evaluation of Phishing Detection Research for Security Needs . . . 15
3.2.10 An Optimized Stacking Ensemble Model for Phishing Websites Detection . . . 16
3.2.11 An Efficient Multistage Phishing Website Detection Model Based On the CASE Feature Framework: Aiming at the Real Web Environment . . . 17
3.2.12 Hybrid Phishing Detection Using Joint Visual and Textual Identity . . . 18
3.2.13 Phishing Detection Using Machine Learning Technique . . . 19
3.2.14 Detection of Phishing Attacks using Visual Similarity Model . . . 20
3.2.15 Smart Phishing Detection in Web Pages using Supervised Deep Learning Classification and Optimization Technique ADAM . . . 21
3.2.16 PDGAN: Phishing Detection With Generative Adversarial Networks . . . 22
3.2.17 Phishing Websites Detection Using a Novel Multipurpose Dataset and Web Technologies Features . . . 23
3.2.18 A Novel Phishing Website Detection Model Based on LightGBM and Domain Name Features . . . 24
3.2.19 Detection Of Phishing Attack Using GAN with RFC . . . 24
3.2.20 Towards Lightweight URL-Based Phishing Detection . . . 25
3.3 Literature Review Summary Table . . . 26
3.4 Conclusion . . . 27

4 Proposed Approach and Methodology 31
4.1 Base Paper Methodology . . . 31
4.1.1 VisualPhishNet . . . 31
4.1.2 Visual Similarity . . . 32
4.2 Proposed Methodology . . . 33
4.2.1 Data Collection . . . 33
4.2.2 Classification . . . 36
4.3 Proposed Model . . . 37

4.3.1 DINAP 37
4.3.2 Features Optimization 38
4.3.3 Evaluation Metrics 39
4.4 Conclusion 39

5 Implementation and Test Cases **40**
5.1 Implementation of Base Model 40
5.2 Experimentation by replacing Image Classifier of Base Model 40
5.3 Experimentation by adding Features Optimization Layers 41
5.4 Implementation of DINAP Model 42
5.5 Conclusion 42

6 Experimental Results and Discussion **43**
6.1 Results of Base Case Implementation 43
6.2 Results by Replacing Image Classifier of Base Model 43
6.3 Results by adding Features Optimization Layers 44
6.4 Results of DINAP 46
6.5 Comparison with VisualPhishNet 46
6.6 Results on VisualPhish Dataset 2.0 47
6.7 Conclusion 47

7 Future Work **48**

List of Figures

2.1 **Industry, Innovation and Infrastructure** 5
2.2 **Peace, Justice and Strong Institutions** 6

4.1 **VisualPhishNet Model** 31
4.2 **Traditional One-to-One Matching VS A Learned Visual Similarity Metric** 32
4.3 **Three Input Tensors make up a Triplet Sample for our model input.** 32
4.4 **Description of VisualPhish Dataset.** 33
4.5 **Visual Representation of VisualPhish Dataset** 33
4.6 **DINAP Model** 37

List of Tables

3.1 **Summary of Literature Review** 28

6.1 **Results of VisualPhishNet Model on VisualPhish Dataset** 43
6.2 **Results by Replacing Image Classifier** 43
6.3 **Results by adding Features Optimization Layers** 44
6.4 **Results of DINAP Model** 46
6.5 **Comparison of DINAP and VisualPhishNet on VisualPhish Dataset** 46
6.6 **Results of DINAP and VisualPhishNet on VisualPhish Dataset 2.0** 47

Chapter 1 Introduction

The Internet has brought with it an incredible number of benefits as well as significant security threats. These threats have caught the attention of governments, businesses, and people all around the world. Phishing stands out among these dangers as a persistent threat that preys on the fundamental trust that supports online interactions.

Phishing is a pervasive cyber threat that encompasses a range of deceptive tactics designed to steal sensitive information or deliver malware. It happens when the attacker pretends to be a legitimate website and tricks the victim into giving away crucial information including credentials. It's imperative to recognize and comprehend the diverse types of phishing attacks to fortify defenses against this malicious menace. Some types of phishing include email phishing, spear phishing, pharming, vishing, smishing, clone phishing, whaling, and angler phishing.

It is more important than ever to strengthen our defenses against phishing as the world struggles with cyber security. Existing tactics have led the road to countering known phishing threats. However, when novel attacks take place, existing solutions and research fall short. These "zero-day" attacks are unique to previous patterns and require novel fixes that go beyond the constraints of existing solutions.

This project aims to expand the capabilities of phishing detection. Our project focuses on the detection and identification of previously undetected phishing strategies in an effort to fill the gap created by conventional measures. This project aims to make a positive impact on both the academic community and the larger field of information security. Our search for solutions is increasingly important as the phishing threat develops, strengthening the cornerstones of dependability and trust in a world that is becoming more interconnected.

1.1 Purpose of this Document

The purpose of this document is the detection & identification of novel attacks in phishing using AI-based algorithms. Novel attacks are one of the most lethal cyber threats which is becoming more dangerous with the passage of time. Numerous new classification-related algorithms and models have been created as a result of advancements in AI. The project will focus on improving the overall accuracy of detecting and identifying novel attacks in web page phishing using visual similarity. The goal of this research is to apply multiple AI based algorithms to detect & identify novel attacks in phishing using visual similarity approach.

1.2 Intended Audience

The intended audience of this research are cyber security professionals who are working on systems used to defend against cyberattacks and the research scholars who are researching in the domain of cyber security. This research will serve as a foundation for both cyber security professionals to develop improved systems that will protect individuals from cyber attacks and researchers to use this as a base for their research in this domain.

1.3 Definitions, Acronyms, and Abbreviations

AI: Artificial Intelligence

ML: Machine Learning

DL: Deep Learning

Novel Attack: Novel attack refers to a newly identified or unidentified cyber security threat that attackers use to exploit vulnerabilities in computer systems or networks.

1.4 At the End of Introduction Chapter

Chapter 1 gave a brief premise of the purpose and inspiration for the project. Chapter 2 gave you a more detailed overview of our vision, more specifically, the scope and domain of our project. Moreover, a problem statement and elaboration were also defined in the chapter. Chapter 3 provided an extended list of literature reviews that had been researched on the domain of phishing detection. As you progress through the report, you will encounter Chapter 4: Proposed Methodology where we detail our approach to tackling the novel phishing detection challenge. Chapter 5: Implementation and Test Cases provides insights into the practical aspects of translating our proposed methodology into a working model, detailing the implementation process and various AI algorithms being tested against our model.. Chapter 6: Experimentation Results and Discussion presents the outcomes of our experiments, critically analyzing the results and engaging in a comprehensive discussion.

Chapter 2 Project Vision

We envision to leverage the strength of numerous deep learning (DL) and machine learning (ML) algorithms to pioneer the identification of novel phishing attacks. The visual-similarity technique is still underutilized because of computational limitations, despite the fact that current research in phishing detection predominantly tackles visual-similarity-based, HTML structure-based, URL-based, or hybrid systems. Our paper aims to close this gap. We seek to further our research by enhancing our dataset with regionally contextualized website screenshots with an emphasis on Pakistani origins. Our model will go through a two-stage process employing sophisticated feature selection techniques and classification algorithms, first classifying phishing and authentic websites, then fine-tuning parameters using incorrectly identified cases. This strategy, which explores novel phishing detection, is in line with our goal of enhancing web security against new threats.

2.1 Problem Domain Overview

Our project aims to detect novel phishing attacks using multiple deep learning (DL) and machine learning (ML) algorithms. Research has already been done in the domain of phishing detection, however, very few of these papers have dealt with novel or zero-day attacks. Typically, there are 4 approaches to phishing detection: visual-similarity-based, HTML structure-based, URL-based, and a hybrid system. Most of these are used frequently, however, a visual-similarity-based approach is not often used. This is mostly because of the computation and dataset constraints faced by research teams considering that this approach extracts features from website screenshots. Nevertheless, our paper aims to use this approach considering the lack of research available using this approach.

Our project firstly aims to supplement the dataset we have procured from previous research using a visual similarity approach [1]. By adding local context to the dataset through the addition of website screenshots of Pakistani origins, we want to make it more robust. Our model will use feature selection techniques and algorithms as well as classification methods. The model will then go through a two-stage method: classification of phishing and legitimate websites, and tuning parameters and weights to train on the wrongly classified examples in the second stage.

A few previous papers have already researched upon detection of novel attacks in phishing and also researched a visual-similarity approach. However, very few papers have been published that use a visual-similarity approach to classify novel phishing websites from legitimate ones.

2.2 Problem Statement

There are very few researches available that provide a mechanism for the detection of novel phishing attacks on websites. The ones that do typically either use a URL-based approach or an HTML structure-based approach. We aim to provide a computationally efficient, highly accurate and fast model that uses a comprehensive dataset of website screenshots that is able to extract features from screenshots of websites and use them to classify novel phishing websites.

2.3 Problem Elaboration

Phishing is the most common cyber security attack currently. Stolen credentials are the most common data breach to occur due to phishing and the attacker has many ways to damage the victim including identity theft, harm to reputation, or damage to their finances. The recent pandemic of Covid-19 demonstrated how much of a threat this is considering more and more people now work remotely. We aim to recognize the threat and provide a mechanism to accurately detect novel phishing attacks on websites to advance the research footprint in this domain.

2.4 Goals and Objectives

Our primary goal is to improve the overall accuracy of detecting and identifying novel attacks in web page phishing. Following are the objectives that we intend to achieve throughout our plan:

- Detection and Identification of Novel Attacks in Phishing using visual attributes:

  Our prime objective would be the detection & identification of novel phishing attacks based on their visual attributes.

- Implementation of Base Case:

  In the beginning, we expect to create an environment that will include a code written in Python and a dataset that has been previously used in the paper, “Zero-Day Phishing Website Detection by Visual Similarity”.

- Enhancing the Dataset:

  Our third objective would be to enhance the dataset that they have used. This will be done by researching what other datasets have been used before and see how we can aggregate these datasets.

- Applying multiple Artificial Intelligence Algorithm(s):

  Our last objective would be to use a range of AI based algorithms to see which of them trains

on the data best and outputs the most accurate and favorable results.

2.5 Project Scope

The project's scope encompasses several key objectives. Our primary aim is to classify novel phishing attacks based on their visual attributes. Our project will encompass web page phishing, while other forms of phishing attacks such as email and personal phishing etc will not fall within the scope of our project. Another significant aspect of our project involves improving the VisualPhish Dataset. We will enhance the dataset by curating additional data, refining labeling accuracy, and ensuring a broader representation of phishing attack variations. Furthermore, we will also do a comprehensive comparative analysis of existing AI based algorithms and models designed for similar phishing attack classification tasks. This evaluation will provide us an understanding of the strengths and limitations of various approaches. If time permits, we aspire to enhance the accuracy of a triplet Convolutional Neural Network (CNN), which is used in the "Zero-Day Phishing Website Detection by Visual Similarity" [1]. This enhancement will involve refining the network's architecture and training methodologies, potentially leading to improved accuracy in classifying phishing attacks.

2.6 Sustainable Development Goal (SDG)

2.6.1 Industry, Innovation and Infrastructure

One of the SDG areas that our research is targeting is Industry, Innovation and Infrastructure. Building resilient infrastructure, fostering innovation, and promoting sustainable industrialization are the primary goals of the Industry, Innovation, and Infrastructure SDGs, as seen in Figure 2.1.



Figure 2.1: Industry, Innovation and Infrastructure

The goal of our research is to bring innovation in the field of cybersecurity which is very important in today's digital world. We hope that our research will help cybersecurity organizations and professionals in developing innovative & sustainable cybersecurity solutions that will protect all of us from cyber-attacks.

2.6.2 Peace, Justice and Strong Institutions

The second SDG area that we will be targeting in our research is Peace, Justice and Strong Institutions. The main objectives of the Peace, Justice, and Strong Institutions SDG, as seen in Figure 2.2, are to promote peaceful and undiscriminating societies for sustainable development, ensure justice for everyone, and construct an accountable, effective, and undiscriminating institution at every level. Cyber attacks have grown increasingly in today's digital world due to global technological advancements. Many individuals, organizations, and institutions have fallen victim to cyberattacks which negatively impact the peace of the society. Our research's goal is to safeguard people from cyberattacks and protect their personal and sensitive information which will ultimately bring peace in society.



Figure 2.2: Peace, Justice and Strong Institutions

Chapter 3 Literature Review / Related Work

This section comprises comprehensive literature reviews of research conducted by researchers for phishing detection, delving into the methodologies and algorithms they used to solve this problem. Additionally, we will present a summarizing table at the conclusion of this section, offering a brief overview of the findings from each paper.

3.1 Definitions, Acronyms, and Abbreviations

CNN: CNN stands for ”Convolutional Neural Network”. It is a deep neural network used typically in computer vision and image recognition tasks.
LSTM: LSTM stands for ”Long Short-Term Memory”. It is a type of recurrent neural network (RNN) architecture.

3.2 Detailed Literature Review

This section provides detailed information about the research that has been done by different researchers on Phishing Detection that will be helpful for our research.

3.2.1 VisualPhishNet: Zero-Day Phishing Website Detection by Visual Similarity

3.2.1.1 Summary of the research item

The purpose of this literature [1] is to evaluate the accuracy of detection of novel phishing attacks in websites. In order to achieve this, the researchers employ a visual similarity technique to compare and assess the similarity between a phishing website and a legitimate one. The framework they developed, VisualPhishNet uses a Triplet Convolutional Neural Network (CNN) to detect phishing in a website. Their detection model is the first to use a deep-learning technique that learns a metric based on visual similarity to compare two websites' screenshots to evaluate phishing. Their dataset, VisualPhish, comprises of 155 trusted websites with 9363 pages, 1195 phishing pages preying on 155 websites (the same 155 trusted websites) and a set of 683 test examples with domains different from the domains in the trusted list. Their model works with two stages, in the first stage, training is executed on all screenshots. They use the second stage to tune the weights by training on the examples that were wrongly classified in the first stage. Using their model, they were able to yield an accuracy of 81% to correctly identify if a website was a phishing website or not. In order to provide further evidence of their system's efficacy, they compared their results with 2 pre-trained CNNs, VGG16 and ResNet50. VGG16 was able to provide an accuracy of 51.32% while ResNet50 yielded an accuracy of 32.21%. These results were much

inferior to the 81% accuracy achieved from VisualPhishNet.

3.2.1.2 Critical analysis of the research item

This paper [1] offers a significant advancement in the use of deep learning techniques in the scope of detection of phishing in websites. The authors developed a Triplet Convolutional Neural Network model. This is the first time that a deep learning model has been used to detect zero-day phishing. They devised a method of a visual similarity metric across screenshots of legitimate websites and phishing webpages targeting those websites instead of the traditional one-to-one matching of each screenshot making the model more robust and superior to previous approaches. A noteworthy discovery was their comprehension of the tactics used by attackers; they realized that adding audible noise to phishing sites would degrade the design, brand or identity of the website. As a result, any website that resembled a real website was labelled as phishing if the similarity exceeded a certain threshold.

The paper's strength lies in its comprehensive dataset, consisting of trusted legitimate websites, phishing webpages, and benign examples used for testing. Intriguingly, the authors employed a unique division, using 60% of the phishing dataset for testing and only 40% for training. This is unlike the norm, typically, a larger part of the dataset is reserved for training and a smaller subset is reserved for testing. The paper's two-stage methodology is noteworthy. All screenshots were trained in the initial stage, and the model was then fine-tuned with examples of misclassification to allow for generalization beyond the trusted-list database. An 81% accuracy in phishing website classification was shown by VisualPhishNet. Analysis in comparison to well-known models like VGG16 and ResNet50 demonstrated its superiority, with VGG16 and ResNet50 achieving accuracies of 51.32% and 32.21%, respectively. It is to be noted that VGG16 and ResNet50 are pre-trained convolutional neural networks just like their local model, VisualPhishNet. The authors do not compare their results with other deep learning algorithms such as generative adversarial networks (GAN) or recurrent neural networks (RNN). A comparative study with other algorithms would have further increased their efficacy. The omission of ensemble algorithms, a technique frequently used for increased accuracy, was noted in the paper. Furthermore, the importance of false positive rates is crucial for classification models. Especially in the domain of phishing, it is dangerous if a phishing webpage is classified as legitimate as this could potentially cause a loss for the victim in the form of their data, identity or monetary. Despite these limitations, VisualPhishNet's innovative use of deep learning techniques marks a significant advancement in the field, laying the foundation for future research in phishing website detection.

3.2.1.3 Relationship to the proposed research work

This research paper [1] is really helpful for our research efforts. The authors demonstrated a visual similarity approach, which we have employed in our research as well. This allows us to take into account the considerations our research must have in order to deploy a similar approach. Moreover, this paper provides a comprehensive dataset, VisualPhish. One of the key goals of our research is to use this dataset and supplement it to create an even better dataset that is more diverse by adding local data points as well. The authors also include their model, VisualPhishNet, using a triple convolutional neural network. This helps us in identifying how good results a CNN can provide and then move on to using different models to yield the best accuracy results.

3.2.2 An Effective Phishing Detection Model Based on Character Level Convolutional Neural Network from URL

3.2.2.1 Summary of the research item

Phishing is a highly malicious form of cyber-attack aimed at tricking individuals into revealing sensitive personal information including credentials and monetary information. Current phishing detection methods primarily depend on source code features. It requires scraping the content of web pages. Despite the fact that Artificial Intelligence algorithms have recently been used to detect phishing, they often struggle with identifying new phishing tactics and require manual feature engineering to work. However, recent strides in the use of deep learning have led to the development of more effective classification algorithms. This research paper [2] suggested an efficient deep learning model for phishing detection on the basis of URL of the website which utilizes the use of character-level CNN. It extracts features such as sequential patterns for quick classification of the URL.

3.2.2.2 Critical analysis of the research item

This research paper [2] proposed a phishing detection model using deep learning that can detect phishing sites using only URL features of the webpage. They used four sets of different features and compared different deep learning algorithms using them. The researchers of this paper used the URL of the webpages for phishing detection, which means that their proposed model relies solely on the URL itself and does not take into account the content of the webpage or contextual information. Attackers can easily craft convincing URLs that may seem legitimate but still lead to malicious phishing sites. They are aware of URL-based detection techniques, and they use numerous evasion techniques to bypass them. Additionally, the proposed model may not be well-suited for detecting novel phishing attacks, where the phishing site is entirely new and hasn't been previously identified.

3.2.2.3 Relationship to the proposed research work

This research paper [2] has provided valuable insights for our consideration into our ongoing research endeavors. The authors demonstrated a model that used deep learning that leverages URL-based features of webpages as a means of phishing detection. While this approach holds merit, it is essential to note that certain critical aspects of phishing detection may have been inadvertently omitted in their analysis. Our research shares a similar goal: detecting phishing threats. But we take a different approach by focusing on the visual aspects of web pages. Furthermore, our research endeavors extend to encompass the detection of novel phishing attacks.

3.2.3 Intelligent Phishing Detection Scheme Using Deep Learning Algorithms

3.2.3.1 Summary of the research item

Phishing attacks have grown increasingly sophisticated due to global technological advancements. Their increasing impact highlights the need for more reliable techniques of detecting these threats and protecting important data of individuals. This research [3] suggested a robust phishing detection solution which uses deep learning techniques and considers both the URL and the content of websites including frames, text and images.

3.2.3.2 Critical analysis of the research item

This research paper [3] proposed a phishing detection model in which they used two models, CNN and LSTM. This way they were able to extract a range of website features to accurately detect phishing websites. The researchers of this paper used URLs and the content of websites which is an effective way to detect phishing. But the proposed system's complexity demands significant resources and it may lead to slower detection. Furthermore, its strict content analysis could result in high false negatives and potentially missing legitimate phishing attempts due to subtle content variations. Additionally, it may struggle to detect novel phishing attacks that haven't been previously identified.

3.2.3.3 Relationship to the proposed research work

This research paper [3] contributes valuable insights to our ongoing research efforts. The authors introduced a deep learning model in which CNN and LSTM were combined to identify various website elements and enhance phishing detection. While this approach is effective, it does come with complexities and resource demands that can slow down the detection process. The insights from this paper serve as essential considerations for our research. Our research aligns with a similar objective: detecting phishing attacks. However, our focus is on proposing a computationally efficient and fast phishing

detection model. We take a similar approach by focusing only on the visual aspects of web pages.

3.2.4 Phishing Detection Using Machine Learning Techniques

3.2.4.1 Summary of the research item

In this research paper [4], the authors conducted a comparative analysis of various models for detecting phishing websites. These methods include Decision Trees, k-nearest neighbors (KNN), ADA Boost, Artificial Neural Networks (ANN), and Support Vector Machine (SVM). The researchers used an extensive set of features that were extracted from the URL string for their model in order to make it more robust and effective.

3.2.4.2 Critical analysis of the research item

In this research paper [4], researchers employed twelve distinct Machine Learning algorithms to analyze a dataset consisting of 6,157 legitimate and 4,898 phishing websites. They focused on 30 features extracted from these websites, particularly from their URLs, which is a promising approach for phishing detection. However, it's important to acknowledge potential challenges. When dealing with numerous features, there's a risk of overfitting, where the model becomes too specialized to the training data. Maintaining a dataset with numerous features can be demanding, requiring continuous updates and potentially lacking coverage for all possible scenarios. Furthermore, a system with a multitude of features may struggle to adapt to rapid changes in the online landscape or handle substantial web traffic effectively. Additionally, they might find it challenging to detect novel phishing attacks that haven't been encountered before.

3.2.4.3 Relationship to the proposed research work

This research paper [4] is really helpful for our ongoing research efforts. The authors did a comparative analysis of twelve different Machine Learning algorithms on a dataset consisting of 30 extracted features. One of the major objectives of our research is also phishing detection and insights from this paper serve as essential considerations for our research. However, we are focusing on the visual aspects of web pages and proposing a computationally efficient model that also detects novel phishing attacks.

3.2.5 A Hybrid Phishing Detection System Using Deep Learning-based URL and Content Analysis

3.2.5.1 Summary of the research item

Phishing is one of the most prominent types of cyberattacks for attackers because they can simply reach a huge audience via social media. The majority of detection mechanisms use URLs because of their rapidness and ease of implementation. However, this technique can overlook some phishing attacks. In this research paper [5], authors proposed a hybrid technique that uses features extracted from URL in level one and in the second level it uses features extracted from content for detection mechanism which improves the accuracy of the model. For the URL-based approach they proposed a model that developed a new model, ”Generative Convolutional Neural Network (GCNN)” that combined Convolutional Neural Network (CNN) and Generative Adversarial Network (GAN). For content-based approach they proposed Deep Neural Network (DNN) model. Their dataset comprised 57 features that were extracted from content for DNN model and 73 features that were identified from URLs for GCNN.

3.2.5.2 Critical analysis of the research item

This research paper [5] proposed a hybrid model for phishing detection. In first level the Generative Convolutional Neural Network is used on features extracted from URL if the model labels it as phishing, the final decision is phishing but if the model labels it as legitimate then Deep Neural Network model is used in level two on features extracted from the content. While this approach is effective, the proposed system is resource-intensive and very complex which may result in longer processing times and increased computational requirements. It is using too many features that were extracted from URL and content of webpages which can result in overfitting, where the model becomes highly specialized to the training dataset but performs poorly on new, unseen data. Lastly, using too many features significantly increases the dimensionality of the data which leads to the ”curse of dimensionality,” where the effectiveness of algorithms decreases as the number of features grows.

3.2.5.3 Relationship to the proposed research work

This research paper [5] contributes valuable insights to our ongoing research efforts. The authors proposed a hybrid model for phishing detection which used Generative Convolutional Neural Networks on URL-based features and Deep Neural Network on content-based features. It provides us with important points we need to consider to be able to successfully propose a computationally efficient and fast novel attack phishing detection model that utilizes the visual aspects of web pages.

3.2.6 High Accuracy Phishing Detection Based on Convolutional Neural Networks

3.2.6.1 Summary of the research item

Individuals and companies throughout the world are now more vulnerable to different cyberattacks due to the rising volume of phishing websites. As a result we require more effective and powerful cyber defense mechanisms for protection from such attacks. In light of the ongoing surge in phishing attacks, there is an increasingly pressing demand for more robust methods of identifying phishing attacks and protecting individuals from them. In this research paper [6], authors proposed an approach for achieving phishing detection with high accuracy through a deep learning model. The proposed system uses CNN for classification to distinguish legitimate sites from phishing sites.

3.2.6.2 Critical analysis of the research item

In this paper [6], the authors have proposed a deep learning model that can detect phishing using features that were extracted from URL and web contents. They extracted a total of 30 features from URL and content of webpages and evaluated the proposed model on a dataset of total 11,055 instances. The researchers used a dataset of features that were extracted from URLs and the content of webpages which is an effective methodology for phishing detection. There are potential obstacles that come with this approach. There is risk of overfitting the model due to dealing with numerous features. Managing such dataset can be resource-intensive and it may not encompass every conceivable scenario. Moreover, this system can encounter challenges when adapting to swift changes in the online environment.

3.2.6.3 Relationship to the proposed research work

This research paper [6] is really helpful for our ongoing research. The authors introduced a deep learning model which uses CNN for phishing detection and was trained on a dataset consisting of 30 extracted features. One of the major goals is also phishing detection and insights from this paper serve as essential considerations for our research. However, we are focusing on the visual aspects of web pages for phishing detection and proposing a computationally and resource-efficient model that also detects novel phishing attacks.

3.2.7 An Intelligent Cyber Security Phishing Detection System using Deep Learning Techniques

3.2.7.1 Summary of the research item

Phishing attacks have become one of the most prominent cyberattacks faced by individuals and businesses in recent times. Through such attacks, the phisher's aim is to illicitly acquire confidential in-

formation from the user with the intent to misuse it. There is a pressing need for advanced phishing detection systems to combat the escalating threat posed by phishing emails, which have been proliferating at an alarming pace in recent years. In response to this threat, the authors of the research paper [7] proposed a email phishing detection model that uses machine learning techniques. It also discusses different strategies phishers use to trick users.

3.2.7.2 Critical analysis of the research item

The authors of the research paper [7] have used seven different supervised classification algorithms on one huge dataset which was a combination of 3 different datasets. They applied Support vector machine, Boosted decision tree, Logistic regression, Locally-deep support vector machine and Decision forest separately on their combined dataset. They used 22 features extracted from the email dataset, which is a promising approach for email phishing detection. However, it's important to acknowledge potential challenges. Data imbalance issues may arise because of how the features are distributed, which can affect the model's performance. Working with a dataset with 22 features may require substantial computational resources. A model developed using a dataset with many features may also have limited generalizability to various phishing scenarios or changing phishing techniques.

3.2.7.3 Relationship to the proposed research work

This research paper [7] contributes valuable insights to our ongoing research. The authors applied 7 different supervised classification models on a dataset that had 22 features to detect email phishing. It provides us with important points we need to consider while doing our research. Our goal is to detect web page phishing using visual similarity which is one of the most frequently attempted phishing attacks.

3.2.8 A Novel Phishing Detection Model Based on Revised Multi objective Evolution Optimization Algorithm and Random Forest

3.2.8.1 Summary of the research item

A novel method for phishing detection is presented in this paper [8]. The authors tried to improve the detection accuracy while reducing the false detection rate of phishing websites by merging an evolutionary optimization and classification algorithm. They built their model using special techniques GA-based techniques that allowed them to optimize their feature selection method. The study confirmed the model's higher accuracy compared to existing approaches through rigorous analysis of phishing and legal website URLs across five public datasets. This study represents a significant development in the field and presents a viable option for effective phishing detection.

3.2.8.2 Critical analysis of the research item

The authors [8] approach from the beginning was unique. Notably, they emphasized the significance of a model's recall rate, observing that end consumers would suffer more severe effects if phishing pages were mistaken for authentic websites if they were surfing those webpages thinking they are legitimate when in reality they are phishing pages. They defined the recall rate as the proportion of correctly detected phishing sites to all phishing sites, and this statistic became a crucial evaluation metric. The selection of pertinent features was efficiently guided by the population initialization strategy they used, which increased the model's accuracy. Randomness was also added via the population state-based adaptive environmental selection. This allowed them to reduce the chances of the algorithm being confined in a local optimum.

Despite the model's success, there are ways to improve it further. The algorithm's time complexity might pose challenges with larger datasets, making it necessary to optimize it further. The authors opted for publicly available datasets, demonstrating the model's efficacy against existing techniques. Their comparative analysis, including models like HS, DTOF-ANN, Kmeans-SVM, SMPD, and EXPOSURE, revealed superior performance in both accuracy and recall rate. This research not only highlighted the crucial aspect of recall rate but also showcased a promising direction for phishing detection, setting new standards in the field.

3.2.8.3 Relationship to the proposed research work

This paper [8] offers invaluable insights for our research. Particularly, the evolutionary algorithm's application to URL features that gives us a perspective on different approaches to the problem. Additionally, it serves as a reference, allows us to reconsider how computationally efficient our algorithm should be. Lastly, this study illuminates the distinction between URL-based approaches and visual-similarity methods.

3.2.9 An In-Depth Benchmarking and Evaluation of Phishing Detection Research for Security Needs

3.2.9.1 Summary of the research item

In "An In-Depth Benchmarking and Evaluation of Phishing Detection Research for Security Needs," [9] the authors introduced PhishBench, a pioneering benchmarking framework. PhishBench incorporates systematic taxonomies for URLs, websites, and emails, which not only enables the development of techniques and features but also makes thorough evaluations possible. This methodology identifies the features necessary for the best detection performance by examining prior literature attributes and using

a variety of datasets. Furthermore, the paper provides a solid framework for future research in phishing detection by delving deeply into an examination of phishing features and strategies, combining cutting-edge tools like AutoML and TPOT.

3.2.9.2 Critical analysis of the research item

The authors [9] conducted in-depth analyses of several existing feature importance frameworks and provided a comparative study that placed their work within the context of earlier efforts. Their PhishBench framework, in particular, is notable for its adaptability, incorporating both cutting-edge and traditional feature extraction methods. PhishBench solves the issue of imbalanced datasets by utilizing automated learning techniques like AutoSKLearn and employing procedures to successfully correct the imbalance.Their extensive taxonomy of features, which includes numerous categories for URLs, websites, and emails, is one of their key advantages. Their strategy uses 14 different classifiers and includes a thorough evaluation procedure using datasets from PhishTank and Alexa. Although, the best accuracy and F1 score were yielded through boosting (ensemble model of decision trees), coming in at 97.61% and 97.12% respectively.

However, a notable drawback lies in the absence of a graphical user interface (GUI), which was present in two of the 3 benchmarking frameworks they compared PhishBench with. No GUI does limit the access to PhishBench, requiring users to have at least a moderate level of programming skills, potentially excluding business users from its use. Despite this limitation, PhishBench's ability to rank and select essential features offers valuable insights into phishing detection methods, making it a valuable asset in the research community.

3.2.9.3 Relationship to the proposed research work

The thorough benchmarking strategy used in this paper [9] is perfectly in line with the objectives of our research. We can improve the effectiveness of our phishing detection model by utilizing their taxonomy of characteristics and varied classifier evaluation. Moreover, the insights gleaned from their study will deepen our understanding and help us optimize our suggested model for improved efficacy and accuracy in phishing detection.

3.2.10 An Optimized Stacking Ensemble Model for Phishing Websites Detection

3.2.10.1 Summary of the research item

The authors of this paper [10] proposed to develop a phishing detection method for websites using an optimized stacking ensemble model. Their aim was to use a Genetic Algorithm (GA) as an optimizer that fine-tunes the weights and the parameters of the Artificial Intelligence algorithms. For this, they

shortlisted a set of classifiers from which they would shortlist the top 3 for their stacking ensemble model. Training and testing of that model was carried out on 3 public datasets from Mendeley. They achieved an accuracy of 98.58% with one of the datasets.

3.2.10.2 Critical analysis of the research item

The authors shortlisted a few classifiers for their experimentation analysis [10]. They included XGBoost, AdaBoost, LightGBM, random forest, bagging, and GradientBoost. They used 3 datasets, one retrieved from UCI and the other two from Mendeley. At first, they trained and tested the classifiers without any optimization techniques applied on the 3 datasets. Among all the classifiers and the datasets, random forest performed the best on the dataset 2 with an accuracy of 98.37% and a recall rate of 98.26%. The authors then applied the GA-based optimization technique and evaluated the classifiers again. And although there was not any effect on the accuracy of random forest, but there was a significant increase in performance of GradientBoost with GA-GradientBoost yielding an accuracy of 98.50% and recall rate of 98.54%. This is a significant increase considering without the GA-based optimization method, the classifier's accuracy and recall rate was 97.67% and 97.76% respectively. Therefore, they ended up using the three best performing classifiers for their ensemble method: GA–GradientBoost, GA–XGBoost, and GA–Bagging. Their final optimized stacking ensemble method was able to achieve an accuracy of 97.16%, 98.58%, and 97.39% for dataset 1, dataset 2, and dataset 3, respectively.

3.2.10.3 Relationship to the proposed research work

The methodology proposed in this paper [10] involving an ensemble model for phishing detection for websites aligns with our research goals. This paper allows us to entertain the possibility of using an ensemble method to stack classifiers for our own model. It also outlines a good framework for weight and parameter optimization. While we do not use a GA-based optimization technique, it does help us scope techniques of optimization.

3.2.11 An Efficient Multistage Phishing Website Detection Model Based On the CASE Feature Framework: Aiming at the Real Web Environment

3.2.11.1 Summary of the research item

In this paper [11], the authors provide elements (Counterfeiting, Affiliation, Stealing, and Evaluation) with an eye towards the actual web environment to ensure selection of the most significant and relevant features. They employ a multi-stage process that includes quick filtering to quickly exclude trustworthy websites and an accurate recognition model after that. This model provides good performance while substantially lowering detection time by balancing the dataset through their framework.

3.2.11.2 Critical analysis of the research item

In this paper [11], the authors offer a reliable method for phishing detection. Their CASE feature framework, which includes features for counterfeiting, affiliation, stealing, and evaluation, addresses the complexity of phishing efforts and simplifies feature selection by categorizing phishing characteristics. They propose a multi-stage model that claims to provide precise and quick detection. Notably, their first stage, whitelist filtering stage effectively filters out the top 10,000 websites which have a visit rate of 87.9%. This takes complexity of O (1) which fulfils the reduced detection time requirement of the proposed model. The model's second stage, Accurate recognition stage comprehensively uses the CASE features and the detection model uses three classifiers, AdaBoost, sequential minimal optimization (SMO) and random forest. The multi-stage model with the CASE features was able to achieve an accuracy of 96.59%.

3.2.11.3 Relationship to the proposed research work

This paper [11] offers invaluable insights for our research. Their work offers important insights by combining related multi-stage tactics and making use of various aspects to develop a robust detection model. Their expert treatment of the imbalance and classification problems is in line with the two-stage detection model that we are using.

3.2.12 Hybrid Phishing Detection Using Joint Visual and Textual Identity

3.2.12.1 Summary of the research item

Phishing attacks have evolved considerably in recent years, which has compelled researchers in this domain to explore new techniques. One of these is the identity-based detection. These algorithms have achieved better accuracy in a changing problem statement where phishing attacks become more common each day. Current identity-based solutions frequently suffer from high false positive rates due to difficulty in generalizing a website's identity. To close this performance gap, this research paper [12] proposed a hybrid technique: identity-based phishing detection that utilizes both visuals of webpage and textual identity.

3.2.12.2 Critical analysis of the research item

To improve accuracy of phishing detection, the authors of this paper [12] extended prior work done for phishing detection based on visual identity using logo. They proposed a hybrid identity-based methodology for phishing detection by combining textual and visual identity discovery components. A webpage that does not contain an input field is considered harmless because there is no way to receive the user's

personal information, model will declare such type of webpages as legitimate but If a webpage contains at least one input field, the processing will continue and model will apply visual and textual identity discovery to detect whether that page is legitimate. While the proposed methodology is effective for phishing detection, it can be challenging to ensure that the textual and visual components align with each other. As phishing strategies evolve, maintaining a hybrid detection system might become more difficult. To maintain the system's effectiveness, updates to both textual and visual components may be required. Moreover, storing and organizing textual and visual data can take up a lot of space, especially if the dataset is huge.

3.2.12.3 Relationship to the proposed research work

This research paper [12] contributes valuable insights to our ongoing research efforts. It proposed a hybrid identity-based phishing detection technique that uses webpage visual and textual identity for phishing detection. It serves as a reference, allowing us to reconsider how computationally and resource-efficient our proposed algorithm and methodology for phishing detection should be.

3.2.13 Phishing Detection Using Machine Learning Technique

3.2.13.1 Summary of the research item

In today's digitized world, our dependence on the internet has become indispensable. We engage in a variety of online activities, such as online buying, banking, and booking. However, the rise of phishing threats poses a significant risk to us. Phishing websites that look identical to legitimate attempts to illicitly obtain sensitive and personal information of an individual like login credentials and credit card details etc. Machine learning-based phishing detection methodologies hinge heavily on accuracy. Many researchers of this domain focused more on enhancing feature proposals and classification models & algorithms, the critical aspect of developing effective feature analysis and selection techniques is sometimes overlooked. Regrettably, the crucial task of crafting efficient feature analysis and selection techniques is at times overlooked. The authors of this paper [13] proposed an effective machine learning-based phishing detection approach by incorporating the Support Vector Machine classifier.

3.2.13.2 Critical analysis of the research item

In this research paper [13], the key features that were used by the researchers for phishing detection include vocabulary, host, and words extracted from the URL. To process these features, each URL is transformed into domain-specific words using a Weka function called "StringtoWordVector." Once they were transformed into vectors, they used a Support Vector Machine (SVM) for building a phishing attack detection classifier. The proposed model relies solely on the features extracted from the URL of

the webpage itself and does not use the content or contextual information of the webpage for phishing detection. Phishers have the ability to skillfully construct URLs that appear legitimate to individuals but ultimately direct them to malicious phishing websites.

3.2.13.3 Relationship to the proposed research work

This research paper [13] has provided us with valuable insights for our consideration in our ongoing research efforts. The authors proposed a machine learning model that used features that were extracted from the URL of webpages for phishing detection. While this approach holds merit, the authors may have inadvertently overlooked some vital aspects of phishing detection in their analysis. Our research shares a similar goal: detecting phishing threats. However, we are focusing on the visual aspects of web pages for phishing detection and proposing a computationally feasible and highly accurate model.

3.2.14 Detection of Phishing Attacks using Visual Similarity Model

3.2.14.1 Summary of the research item

The authors of this research [14] presented a robust system that consists of two parts, classification and phishing detection. The proposed approach successfully recognizes phishing attempts by inspecting website URLs and structures. The solution offers a comprehensive strategy to identify phishing assaults, demonstrating its effectiveness in increasing web security by using features selection, WHOIS phishing databases, and blacklists.

3.2.14.2 Critical analysis of the research item

In this paper [14], the authors employ a structured two-stage approach, involving classification and detection. Their methodology utilizes a dataset comprising both legitimate and phishing websites. The system has three sub-modules: data gathering, feature selection, and classification. The thorough feature extraction process takes into account a number of factors, including the URL bar, unusual features, HTML & Java, and domain-based characteristics. The model delivers decent results, with an accuracy of 86.02 percent and a low false negative rate of 1.48 percent, using a Support Vector Machine (SVM) as the classification technique. This strong performance highlights the model's efficiency in correctly classifying phishing assaults.

3.2.14.3 Relationship to the proposed research work

Unlike other papers, the authors [14] were very particular in defining the problem at hand and its scope. This has helped us understand the problem better. The use of SVM classification and several feature

extraction techniques has provided us the opportunity to entertain SVM as a potential algorithm to use in our work.

3.2.15 Smart Phishing Detection in Web Pages using Supervised Deep Learning Classification and Optimization Technique ADAM

3.2.15.1 Summary of the research item

Phishing is the process of misrepresenting malicious websites as legitimate ones which is done by attackers to gain sensitive and personal information from the individuals. It is regarded as one of the most serious threats in cybersecurity. The majority of phishing detection methodologies which are currently used employ bayesian classification to distinguish phishing webpages from legitimate webpages. These techniques offer accuracy of up to 90% but with datasets which have fewer webpages. Current methodologies have not been accurate enough for massive datasets. This research paper [15] proposed a new approach that used hyperlinks found in the website's HTML page's source code for phishing detection. It uses 30 features in training the supervised Deep Neural Network model with the Adam optimizer to distinguish between legitimate and phishing websites.

3.2.15.2 Relationship to the proposed research work

This research paper [15] has provided us with valuable insights for our consideration into our ongoing research. The authors proposed a deep learning classification model that uses features extracted from the URL of webpages for phishing detection. While this methodology is good for phishing detection, it may not be effective and accurate. Our research also has a similar goal: detecting phishing attacks. But we take a different approach by focusing on the visual aspects of web pages and proposing a computationally feasible and highly accurate model.

3.2.15.3 Critical analysis of the research item

In this research paper [15], authors proposed a methodology that uses a feature vector with 30 parameters for phishing detection. The suggested deep learning model with Adam Optimizer implements a Listwise approach to distinguish between phishing and legitimate websites. They have used 30 heuristics training their proposed model that were mainly extracted from the URLs which means that their proposed model relies solely on the features extracted from the URL and does not use the contextual and visual information of the webpage. Phishers can easily craft such URLs that may seem legitimate but still lead to malicious phishing sites. Additionally, the proposed model may not be able to detect novel phishing attacks.

3.2.16 PDGAN: Phishing Detection With Generative Adversarial Networks

3.2.16.1 Summary of the research item

Phishing is one of the most dangerous cyberattacks. It has caused many financial losses and identity thefts. Due to the increase in online electronic services and payment systems there is a need for a highly-accurate phishing detection system which can detect and protect individuals from phishing attacks. Deep learning has gained popularity recently as a method for identifying malicious phishing websites. The authors of this research paper [16] proposed a phishing detection model known as PDGAN which exclusively relies on features extracted from a website's URL for phishing detection. It employs a combination of LSTM and CNN. While CNN acted as a discriminator, checking whether the URLs are indicative of phishing or legitimate websites, LSTM served as a generator, synthesizing phishing URLs.

3.2.16.2 Critical analysis of the research item

Authors of this paper [16] proposed a deep learning model which was a combination of CNN and LSTM models for phishing detection using URLs of webpages only. In the first step, LSTM was used to generate synthetic phishing URLs, and after that CNN was used to check whether the URL is phishing or legitimate. While this approach holds its merit, the authors only used features that were extracted from the URL of webpages and they didn't use any visual and contextual features of that webpages. Phishers can easily bypass such URL-based detection mechanisms and can easily create convincing URLs that may seem legitimate to an individual but still lead to malicious phishing sites. Additionally, the proposed model may not be able to detect novel phishing attacks, where the phishing site is entirely new and hasn't been previously identified.

3.2.16.3 Relationship to the proposed research work

This research paper [16] is really helpful for our ongoing research. The authors introduced a deep learning model which used both LSTM and CNN for phishing detection and was trained on a dataset that consisted of features of URL. One of the major goals is also phishing detection and insights from this paper serve as essential considerations for our research. However, we are focusing on the visual aspects of web pages for phishing detection and proposing a computationally and resource-efficient model that also detects novel phishing attacks

3.2.17 Phishing Websites Detection Using a Novel Multipurpose Dataset and Web Technologies Features

3.2.17.1 Summary of the research item

In this study [17], they present a novel method for phishing detection in websites. The authors created the Phishing Index Login Websites Dataset (PILWD), which contains 134,000 confirmed samples, by utilizing URL, HTML, and web technology elements. They extracted a set of features from web technologies by browsing reliable login websites to produce an extensive reference. When tested on the PILWD dataset, their model had a remarkable 97.95% accuracy utilizing a LightGBM classifier.

3.2.17.2 Critical analysis of the research item

In the paper [17], "Phishing websites detection using a novel multipurpose dataset and web technologies features," the authors ensure a balanced representation of homepages and login webpages by building two distinct legitimate subsets. The detection model is enhanced by their addition of innovative features, including technology analysis and distinctive HTML attributes, in addition to traditional features. The research suggests 27 individually created features, each of which contains crucial data for precise detection. The curation of the Phishing Index Login Websites Dataset (PILWD-134K) represents a significant advancement by providing adaptable features for various phishing detection techniques. Their dataset includes samples of URL, HTML, screenshots of the website, technology analysis and an offline version of the website. Due to its data-intensive nature, combining numerous data points from a single website into a single sample, however, gives computational difficulties. The study uses a variety of classifiers, including Adaboost (ADA), Support Vector Machine (SVM), Logistic Regression (LR), k-Nearest Neighbour (kNN), Naive Bayes (NB), Random Forest (RF), LightGBM (LGBM), and XGBoost (XGB). Notably, LightGBM performs the best at 97.95% accuracy. Although the study's large dataset increases accuracy, the sheer number of samples also adds the risk of isolated false positives or negatives.

3.2.17.3 Relationship to the proposed research work

This project has benefited greatly from this research [17]. Their feature addition and approaches to dataset expansion are one of the goals that we aim to achieve too. The proposed work's investigation of machine learning methods is in line with the use of several classifiers, including LightGBM. Their difficulties with handling large amounts of data, however, identify prospective difficulties and allows us to prepare us to avoid that mistake.

3.2.18 A Novel Phishing Website Detection Model Based on LightGBM and Domain Name Features

3.2.18.1 Summary of the research item

The authors of this study [18] present a novel phishing detection model. The study suggests a method to recognize phishing websites by utilizing domain name attributes and LightGBM, a machine learning algorithm. This model provides a solid approach for precise and effective phishing detection by first extracting domain name features and then using LightGBM for classification.

3.2.18.2 Critical analysis of the research item

The authors [18] highlight the difficulties that current methods have in managing the real-time identification of a large number of domain names and provide a novel method for phishing website detection in their study that can be deployed for real-time detection. The major steps in the model's methodology are domain name feature extraction, domain name feature selection, relevant feature selection, and using LightGBM as the classifier. The model obtained an accuracy of 93.88% using publicly accessible datasets from Alexa and PhishTank. However, there are some limitations. The model's training and testing relies on existing datasets, despite the claim that it can identify novel phishing attacks. As a result, the novelty of the detected attacks is called into question because the model hasn't been validated properly. Despite the method's ability to handle complicated domain name properties, testing the model against genuinely innovative and previously unheard-of threats is still necessary to confirm the model's effectiveness in real-world circumstances.

3.2.18.3 Relationship to the proposed research work

This research [18] aligns closely with the proposed work. Their use of domain name features and LightGBM resonates with our approach, emphasizing the importance of using different classifiers for training. However, their model's challenge in truly detecting novel attacks is different from our paper, which focuses on genuine zero-day threats, ensuring robust cybersecurity measures against novel attacks.

3.2.19 Detection Of Phishing Attack Using GAN with RFC

3.2.19.1 Summary of the research item

The study [19] outlines a technique that combines Random Forest Classifier (RFC) and Generative Adversarial Networks (GANs). GAN creates realistic phishing samples, improving the training data. The RFC ensures accurate classification of phishing websites by examining relevant features. The accuracy

and practical usefulness of the model are considerably increased by this novel combination of GAN and RFC approaches.

3.2.19.2 Critical analysis of the research item

In their study [19], the researchers introduced a phishing detection model by combining the Random Forest Classifier (RFC) with Generative Adversarial Networks (GAN), focusing on URL analysis. Their approach, dives into the intricacies of Feature Selection, Algorithm Tuning, and dataset augmentation. Thirty crucial features are analyzed to assess website legitimacy. Leveraging the GAN's generator component, they produce synthetic URLs, supplementing the training dataset. By training the GAN on a mix of genuine and phishing URLs, they create a comprehensive dataset. This combined real and synthetic data undergoes detailed feature extraction and preprocessing, refining the dataset for the RFC model. Their model exhibits an accuracy of 97.58%. The researchers assert its potential deployment in real-time phishing detection and URL-based classification environments, making a compelling case for its practical application.

3.2.19.3 Relationship to the proposed research work

This research [19] aligns closely with the proposed work, enhancing phishing detection through innovative methods. While our focus is on zero-day attacks, their approach to augmenting datasets and optimizing classifiers resonates with our aim to improve model accuracy. Their successful augmentation of their dataset is impressive and is something that we will take notes from.

3.2.20 Towards Lightweight URL-Based Phishing Detection

3.2.20.1 Summary of the research item

Mostly used computing devices use web browsers to access information and online services. There are safety risks associated with the widespread use of web browsing in our daily lives because the general public lack situational awareness of cyberattacks. As a result, they are vulnerable to a range of threats, including ransomware, phishing and malware etc. Phishing attempts in the present threat landscape usually function as the first stage of a more elaborate operation. Research has shown that deny lists cannot shield individuals from the dynamic phishing URLs. In this research paper [20], authors proposed a supervised machine learning algorithm which used a combination of features that are extracted solely from the URL to detect and block phishing attacks.

3.2.20.2 Critical analysis of the research item

Authors of this research paper [20] trained and evaluated on several machine learning algorithms, including Naive Bayes, Decision Tree, Random Forest, Support Vector Machine, and Multi-Layer Perceptron in order to propose the algorithm that worked best for phishing detection engine on their dataset which consisted of 12 features that were extracted from URL. Ultimately, they opted for the Random Forest algorithm for detection of phishing attacks on their dataset and were able to achieve good results. The proposed approach is good for phishing detection but they didn't use any visual and HTML features of the page and only used dataset which was extracted purely from the URL. Phishers are aware of such URL-based detection techniques and can easily craft such convincing URLs that may seem legitimate to an individual but still lead to malicious phishing sites. Additionally, the proposed model may not be able to detect novel phishing attacks, where the phishing site is entirely new and hasn't been previously identified.

3.2.20.3 Relationship to the proposed research work

This research paper [20] contributes valuable insights to our ongoing research efforts. The authors introduced a supervised machine learning algorithm in which they used Random Forest to identify phishing detection using features that were extracted from the URL. While this approach holds its merit, it does not take into account the visual aspects of the webpage which makes the proposed approach less effective. Our research aligns with a similar objective: detecting phishing attacks. However, our focus is on proposing a computationally efficient and fast phishing detection model by focusing only on the visual aspects of web pages.

3.3 Literature Review Summary Table

During the course of our research, we studied many existing papers published in the domain of phishing detection. We observed that a range of techniques and methodologies were employed in them. You will find the name of the author, the method used, the results yielded and the limitations they faced in Table 3.1.

3.4 Conclusion

We have read the methodologies and techniques of the different papers that researched phishing detection. Typically used models were classifiers including CNN, SVM, Bagging, Boosting and LSTM. Only a few of the research done detect novel phishing attacks. An even smaller number of these researchers detect novel attacks in phishing using a visual-similarity approach.

Table 3.1: Summary of Literature Review

The summary of various research that has been done for phishing detection in the past from 2020-2023 is presented here.

Author	Method	Results	Limitations
Sahar et al. [1]	Used Triplet CNN on a visual dataset that comprised of 9363 legitimate pages and 1195 phishing pages from 155 websites.	Were able to achieve an accuracy of 81%	Proposed Model is resource-intensive
Aljofey et al. [2]	Applied a deep learning-based phishing detection model that used only features extracted from URL of the webpage for phishing detection.	Were able to achieve an accuracy of 95.02%	Model does not take content of website into account and only relies on the features extracted from URL
Adebowale et al. [3]	Proposed phishing detection solution which uses combination of 2 deep learning algorithms CNN & LSTM and uses both the URL and the content of the webpage.	Were able to achieve an accuracy of 93.28%	Proposed system is complex and resource-intensive.
Shahrivari et al. [4]	Applied 12 distinct Machine Learning algorithms to analyze a dataset of 30 features extracted from these websites, particularly from their URLs.	Were able to achieve an accuracy of 97% on XGBoost	Risk of overfitting because it's dealing with numerous features and maintaining such dataset can be demanding, potentially lacking coverage for all possible scenarios.
Korkmaz et al. [5]	Proposed a model by combining the Generative Convolutional Neural Network for URL-based approach and a Deep Neural Network model for content-based approach.	Were able to accuracy of 98.37%	Proposed system is very resource-intensive and very complex which may result in a longer time for phishing detection.
Suleiman et al. [6]	Proposed a deep learning model that uses CNN to detect phishing using features that were extracted from URL and web contents.	Were able to achieve an accuracy of 98.2%	Proposed system can encounter challenges when adapting to swift changes in the online environment and risk of overfitting the model due to dealing with numerous features.
Mughaid et al. [7]	Applied seven different supervised classification algorithms on one huge dataset which was a combination of 3 different dataset.	Were able to achieve an accuracy of 96%	Proposed model was developed using a dataset with many features that may also have limited generalizability to various phishing scenarios plus Data imbalance issues may arise because of how the features are distributed.

Zhu et al. [8]	Used a Revised Multi-Objective Evolution Optimization Algorithm (MOE) and Random Forest (RF) on 5 different datasets.	Were able to achieve an accuracy of 95.92%	Does not use a curated dataset
Ayman et al. [9]	Developed a new benchmarking framework. Included 14 different classifiers in their framework, Boosting yielded the best results.	Were able to achieve 97.61%	Their benchmark does not have a GUI and is difficult to use.
Al-Sarem et al. [10]	Used a GA-based optimization technique and XGBoost, AdaBoost, LightGBM, Random Forest, Bagging, and GradientBoost as classifiers.	Were able to achieve an accuracy of 98.58%	Small sample size
Dong-Jie et al. [11]	Developed a CASE feature framework for feature selection and use AdaBoost, sequential minimal optimization (SMO), and random forest.	Were able to achieve an accuracy of 96.59%	Their model is a resource-intensive and complex model with multiple stages.
Tan-Jie et al. [12]	Proposed a new hybrid identity-based phishing detection technique that used both visuals of webpage and textual identity.	Were able to achieve an accuracy of 98.6%	Maintaining a hybrid detection system might become more difficult and it can be challenging to ensure that the textual and visual components align with each other.
Rashid et al. [13]	Proposed an effective machine learning-based phishing detection approach by using the Support Vector Machine classifier.	Were able to achieve an accuracy of 95.66%	Model relies solely on the features extracted from the URL of the webpage and does not use the contextual information
Paturi et al. [14]	Their model analyzed the URL and structure of the website and used SVM as a classifier.	Were able to achieve an accuracy of 86.02%	They do not have an optimized feature selection technique
Santhaiah et al. [15]	Proposed a deep learning-based phishing detection model with the Adam optimizer that used only features extracted from URL of the webpage for phishing detection.	Were able to achieve an accuracy of 98%	Proposed model solely relies on the URLbased-features and does not use the visual information of the webpage.

Saad et al. [16]	proposed a phishing detection model known as PDGAN which uses a combination of LSTM and CNN and exclusively relies on features extracted from a website's URL for phishing detection.	Were able to achieve an accuracy of 97.58%	Proposed model solely relies on the URLbased-features and does not use the visual information of the webpage.
Manuel et al. [17]	Used a variety of classifiers on the dataset they built with each sample consisting of URL, HTML, screenshots, offline website.	Were able to achieve an accuracy of 97.95%	The large number of samples in the dataset adds risk of false negatives and false positives.
Zhou et al. [18]	Used lightGBM as classifier on domain name features for phishing detection.	Were able to achieve an accuracy of 93.88%	The authors claim their model to detect novel attacks despite it not being tested against novel data.
Gayathri et al. [19]	Used GAN to supplement their dataset and RFC to classify legitimate and phishing samples.	Were able to achieve an accuracy of 97.58%	Their scope of URL-only features limits their model
Andrei et al. [20]	Proposed supervised machine learning algorithm Random Forest which used a combination of features that are extracted solely from the URL to detect and block phishing attacks.	Were able to achieve an accuracy of 98.86%	Proposed model solely relies on the URLbased-features and does not use the visual information of the webpage.

Chapter 4 Proposed Approach and Methodology

One of the biggest challenges in the field of cybersecurity is the threat posed by previously unknown or "novel" phishing attempts on websites. There are very few tools available that are intended to identify new phishing attempts. Previous research has mostly focused on studying the URL or HTML structure of a webpage for detecting phishing, frequently ignoring the possibility of visual similarity as an identification technique. Our study aims to close this gap by presenting a fast, highly accurate, and computationally efficient model. Our approach leverages a large dataset of website screenshots to extract features for the classification of new phishing websites—a dimension that has not been well studied. Recognizing the limitations of existing models, often characterized by computational intensity, our approach strives to address the critical need for a pragmatic and effective solution to novel phishing detection.

4.1 Base Paper Methodology

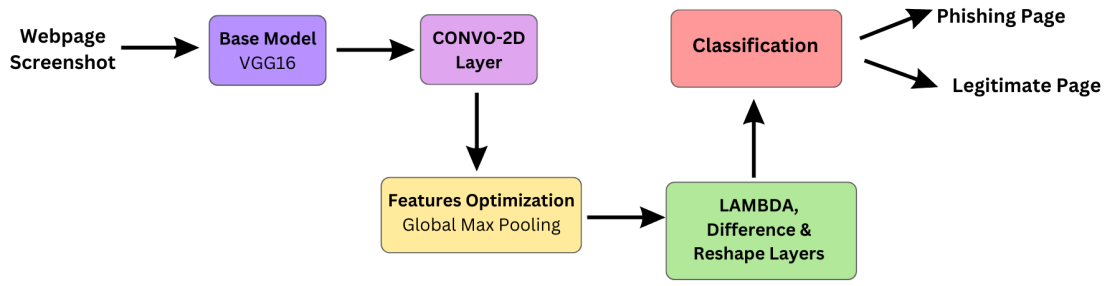


Figure 4.1: VisualPhishNet Model

4.1.1 VisualPhishNet

VisualPhishNet [1] as shown in Figure 4.1, uses a Triplet Convolutional Neural Network (CNN) to detect phishing in a website. It uses VGG16 as the base model. A new convolutional layer of Convo-2D is added on top of the base model. After that, a global max-pooling layer is applied. Two Lambda layers are used. One lambda layer computes the mean along the axis of feature vectors, and the other lambda layer computes the squared difference between the anchor-positive and anchor-negative encodings. After that there is a difference layer in which the squared differences are subtracted to obtain a single value which represents the difference between the anchor-positive and anchor-negative feature vectors. Lastly, the resulting difference is reshaped to have a single dimension using Reshape Layer.

4.1.2 Visual Similarity

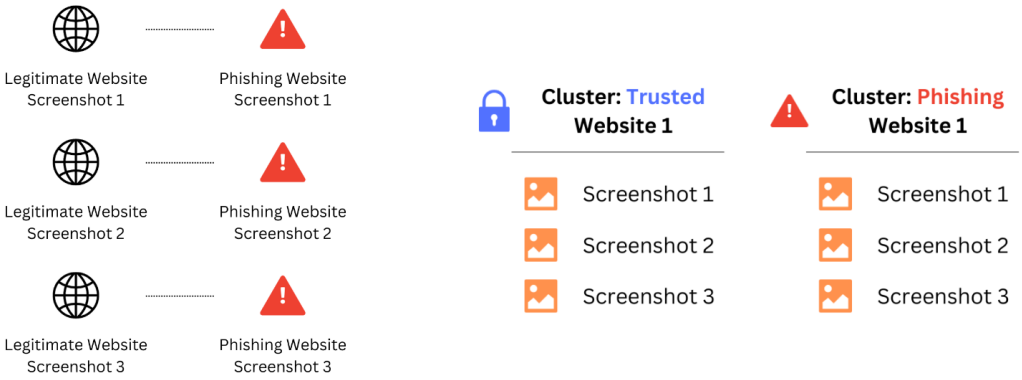


Figure 4.2: Traditional One-to-One Matching VS A Learned Visual Similarity Metric

VisualPhishNet is the first to use a deep-learning technique that learns a metric based on visual similarity to compare two websites' screenshots to evaluate phishing. This is different from a traditional one-to-one matching between a phish and a trusted webpage, this comparison is demonstrated in figure 4.2. According to their research, the visual similarity metric used can discriminate between legitimate and phishing webpages. Three input tensors, each representing an image converted into a 224x224x3 matrix, are taken as input. The anchor image, picked from the legitimate subset and from a random cluster such as Microsoft, Google, or JP Morgan, acts as the reference point. A positive image, picked from the same cluster as the anchor image but with an 85% probability of being chosen from the legitimate set, and a negative image from a different cluster from the anchor, contribute to the model's learning process. The model calculates the distance between the anchor image and the positive image, let's call this distAP, and the distance between the anchor image and the negative image, let's call this distAN. A margin is imposed that conditions that the absolute difference between distAP and distAN should be greater than or equal to the margin. This allows the model to learn and then later determine if 2 webpages are similar or not. Visual representation is shown in Figure 4.3

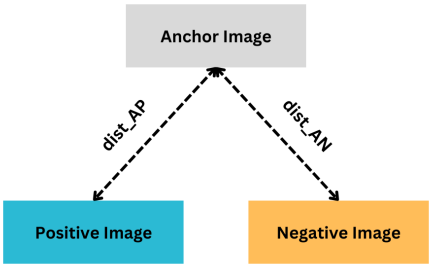


Figure 4.3: Three Input Tensors make up a Triplet Sample for our model input.

4.2 Proposed Methodology

This sections provides the detailed procedure that we followed to propose our DINAP model.

4.2.1 Data Collection

4.2.1.1 Dataset Description

Trusted Pages	9,363
Phishing Pages	1,195
Benign Pages	683

Figure 4.4: Description of VisualPhish Dataset.

The dataset used in this study was curated and first used by Abdelnabi et al [1] and consists of a large collection of screenshots of websites, including both phishing and trusted websites. A visual representation of our dataset is depicted in Figure 4.5 It offers a representative sample of authentic web pages with a thorough selection of 9,363 pages taken from 155 legitimate domains. The collection also includes a subset of 1,195 phishing samples that were taken from PhishTank to target the same 155 trusted websites. In order to bolster the dataset's resilience even more, a supplementary subset of 683 benign samples was obtained from the highest-ranked websites. This varied assortment guarantees a comprehensive representation of the internet, covering both trusted and harmful sources. In order to develop new, effective, and economical cyber threat mitigation measures, the design and structure of the dataset offer a basis that is conducive to the research and improvement of phishing detection methodologies.

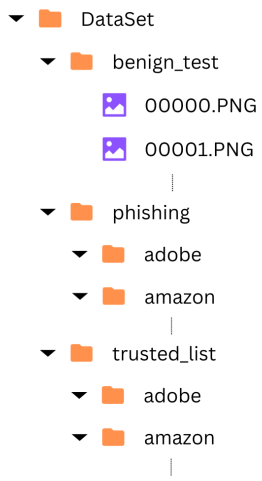


Figure 4.5: Visual Representation of VisualPhish Dataset

4.2.1.2 Data Augmentation

We thoroughly imagined our approach to include a proper search on PhishTank for reported phishing pages of websites belonging to well-known international brands. As soon as we recognised these phishing attempts, we took screenshots of those phishing instances. Using this approach, we were able to get screenshots of both legitimate and phishing webpages, which enhanced our dataset by providing direct comparisons that were essential for training our model.

Our strategy ensures a comprehensive representation of web content, including the differences and similarities between legitimate and phishing webpages. By pairing each legitimate website with its phishing counterpart, we facilitate the model's learning of a robust visual similarity metric. This pairing is pivotal in teaching the model to discern subtle visual cues that differentiate genuine webpages from fraudulent ones.

Our methodology is complex because of the way the dataset is organized. Unlike a one-to-one matching paradigm, it offers a wide range of authentic and fraudulent cases. This diversity greatly improves the model's adaptability and precision by enabling it to detect phishing attempts across a range of pages from the same website. As a result, our dataset augmentation procedure strengthens the model's capacity to identify and thwart phishing attempts with increased precision and efficacy while simultaneously increasing the dataset's applicability to real-world circumstances. We named it VisualPhish 2.0 because newly collected data was merged with original VisualPhish dataset [1] bringing the total number of brands in the dataset to 174.

Dataset Augmentation Strategy: We focused on gathering website screenshots that more correctly depict the phishing landscape. We turned our attention to well-known international businesses that have a larger footprint and a higher chance of being the target of phishing attempts. This is how we methodically carried out this plan:

Sourcing Phishing Examples: We leveraged PhishTank, a reputable resource for identifying reported phishing pages. By filtering for well-known global brands, we were able to retrieve a rich selection of phishing attempts targeting these entities.

Brand-Specific Collection: For each selected brand, we meticulously reviewed all reported phishing examples. This allowed us to gather a comprehensive understanding of the phishing tactics employed against each brand. We have expanded the VisualPhish Dataset by adding 19 brands, bringing the total

number of brands in the dataset to 174, up from the previous 155 brands. The following brands have been included:

- Bank of America
- BBVA Compass
- Binance
- Citi Bank
- Delta Air
- Deutesh Bank
- Discover Bank
- Fifth-Third Bank
- Github
- Halifax
- Intesa Sanpaolo
- Key Bank
- Kiwi Bank
- M&T Bank
- O2
- Tesco
- US Bank
- Vergin Media
- Xfinity

Legitimate Website Screenshots: We started by accessing the official home pages of the corresponding legitimate websites for each brand identified in the phishing examples. This served as our base for further exploration.

Hyperlinked Pages Capture: Beyond the home page, we extended our crawling to include pages directly linked from the home page. This allowed us to capture a broader representation of the website's legitimate interface.

Focus on Critical Pages: Given the significance of login, registration, and sign-up pages in phishing attempts, special attention was directed towards these pages. We specifically crawled and captured screenshots of these critical pages, recognizing their high value for training our detection model.

Dataset Enrichment: By adopting this strategy, we significantly enriched the dataset with a diverse array of legitimate and phishing screenshots. This not only expanded the dataset's coverage but also its depth in terms of the variety of pages and phishing tactics represented.
The inclusion of pages linked from the home page, alongside those of critical significance such as login and sign-up pages, ensured our dataset mirrors the real-world scenarios users encounter. This enhances the model's ability to discern legitimate from phishing content accurately.

4.2.2 Classification

We used a selection of pre-trained image classification models and various versions of those models from the Keras Library. Our objective was to assess the performance of these models on VisualPhish dataset and identify the most suitable image classifier for our model. We prioritized models that are computationally and storage-wise lightweight and have high accuracy in delivering optimal results. Models are listed below:

- VGG19
- EfficientNet
- DenseNet
- Xception
- Inception
- ResNet
- NASNet
- MobileNet
- InceptionResNet

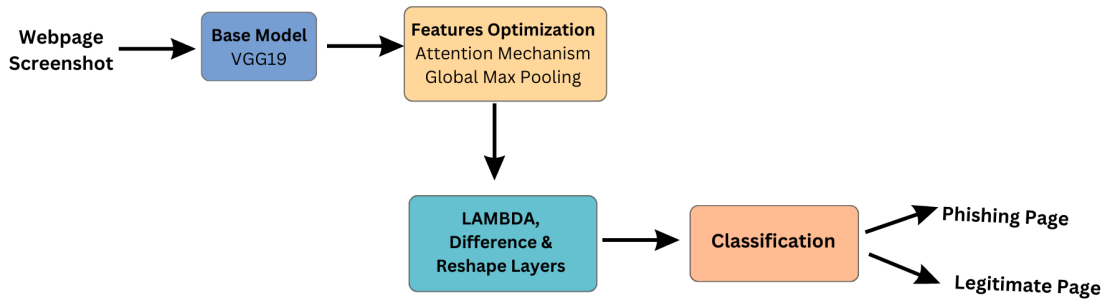


Figure 4.6: DINAP Model

4.3 Proposed Model

4.3.1 DINAP

The proposed model DINAP as shown in Figure 4.6, is a Triplet Convolutional Neural Network(CNN). It includes three input tensors, an anchor input, a positive input, and a negative input. These are carefully defined in the proposed DINAP model to represent anchor, positive, and negative images, respectively.

4.3.1.1 Base Model

It uses VGG19 as a base model, a pre-trained convolutional neural network that has been extensively studied and trained on ImageNet. It consists of 25 layers (16 convolution layers, 3 Fully connected layers, 5 MaxPool layers and 1 SoftMax layer).

4.3.1.2 Lambda layers

The model employs two Lambda layers. The first lambda layer computes the mean along the axis of feature vectors, while the second lambda layer computes the squared difference between the anchor-positive and anchor-negative pairs' encodings.

4.3.1.3 Difference Layer

The squared differences are then subtracted using a Difference Layer, providing a single value that represents the difference between the anchor-positive and anchor-negative feature vectors.

4.3.1.4 Reshape Layer

Resulting difference is reshaped by a Reshape Layer into a single-dimensional representation.

4.3.1.5 Classification

The last classification layers are removed to make place for the installation of a new convolutional layer with specific parameters, which is tailored for the triplet structure. As the encoding process progresses through the base model and the middle layers of the network, feature vectors (encodings) for the anchor, positive, and negative images are produced as the output. Intermediate layers are essential to the architecture of the model since they help get these encodings ready for classification at the end of the model. The robust design of the DINAP model, which combines pre-trained and custom layers, enables it to extract key properties for distinguishing between phishing and legitimate samples.

4.3.2 Features Optimization

During the feature optimization, the model is customized in a subtle way to maximize its ability to identify phishing attempts in the context of image classification. After the base model, additional layers are added that are specifically designed to handle the unique problems that arise from image classification. Some of features optimization layers are:

- Batch Normalization
- Dropout
- Attention
- Global Max Pooling
- Global Average Pooling
- Global Min Pooling

4.3.2.1 Attention

We strategically incorporated an attention mechanism layer to the output of base model's feature vectors. This mechanism enables the neural network with the capability to focus on specific segments of the feature space during the generation of output, making it sensitive to relevant features.

4.3.2.2 Global Max Pooling

The inclusion of a global max pooling layer is a critical component that improves the model's capacity to recognize complex local patterns like logos that are essential for accurate classification.

4.3.3 Evaluation Metrics

Evaluation metrics play a crucial role in designing a deep learning model. Design and architecture can go only so far in this procedure, instead, it is an iterative process of gauging the performance of different techniques with the help of the evaluation metrics identified.

4.3.3.1 Precision

Precision is critical in our context, as the model relies on phishing pages' visual resemblance to trusted websites. We evaluate the model's performance by calculating the correct match percentage between a phishing page and its targeted website, which demonstrates how well it can identify.

4.3.3.2 Accuracy & ROC

The binary classification between legitimate and phishing pages is subjected to a thorough analysis, with overall accuracy another one of the metrics we use to evaluate the model. This multifaceted evaluation incorporates the Receiver Operating Characteristic (ROC) curve area, providing a comprehensive overview of the model's capabilities.

We are still under the process of evaluating the performance of our model with different optimization algorithms applied to the feature vectors. These feature vectors are inputted to the classifier. For the classification layer, we aim to test a range of classifiers with the intention to draw out the best algorithms that suit well to our problem area.

4.4 Conclusion

In conclusion, this chapter provides a comprehensive overview of our proposed methodology, which encompasses key components such as Data Collection, Classification, the structure of the Proposed Model with its distinct layers, and the Evaluation Metrics employed. The methodology is designed to address the difficult challenges posed by novel phishing attacks, employing a visual-similarity-based approach. As we delve into the depth of data collection, the robust classification process, and the sophisticated architecture of our proposed model, the chapter offers a roadmap for navigating the complexities of phishing detection.

Chapter 5 Implementation and Test Cases

In this chapter, we will discuss the implementation details of our proposed model DINAP. It contains details about all the experimentation which was done by us for the implementation of our model DINAP. We'll discuss all the experimentation conducted during the designing implementation of DINAP, providing a comprehensive overview of our methodology in this chapter.

5.1 Implementation of Base Model

Authors of VisualPhishNet [1] have written an authority on novel phishing detection. Their method is considered to be unique yet effective as they were successful in using a visual similarity approach to solve this unique problem. As we began, our first step towards solving this problem was implementing the base model (VisualPhishNet) [1]. We accomplished this by first using the open-source repository of their code files posted on GitHub by the authors of VisualPhishNet [1]. This was followed by acquiring their dataset, VisualPhish which we used to train and test the model on. Moreover, we set up the environment for our experimentation implementation using the Kaggle Notebook. This not only allowed us to test out and experiment with their model but to later build upon their existing methodology, setting the stage for our subsequent advancements.

5.2 Experimentation by replacing Image Classifier of Base Model

As experimentation and comprehension of base paper and model came to an end, it was time to move on to our own implementation. Every image classification problem requires a model that is responsible for feature extraction before the model can move onto any pre-processing or embedding techniques. VGG16 is used as the base model in ViusalPhishNet which in itself is very effective and has provided results for the authors. However, there is a range of image classifiers available to be applied that have shown promising results to image extraction sub-problems especially in the context of website design or logos.

This paper used a selection of pre-trained image classification models and various versions of those models from the Keras Library for experimentation. The goal was to assess the performance of these models on VisualPhish dataset and identify the most suitable image classifier that would act as base image classifier in DINAP's implementation. We prioritized models that are computationally and storage-wise lightweight and have high accuracy in delivering optimal results. Models which we used for our experimentation were:

- DenseNet201

- EfficientnetB0
- EfficientNetV2B3
- EfficientNetV2S
- InceptionResNetV2
- InceptionV3
- MobileNetV2
- NASNetLarge
- NASNetMobile
- ResNet152V2
- VGG19
- Xception

5.3 Experimentation by adding Features Optimization Layers

After fully experimenting with different base image classifiers, we proceeded to the next part of our experimentation and observation which was customizing the middle feature optimization layers. After the base model, additional layers are added that are specifically designed to handle the unique problems that arise from image classification. We required techniques that improved the model's capacity to recognize complex local patterns like logos that are essential for accurate classification. Furthermore, to tackle the problem through visual similarity, the system needed a mechanism that had the capability to focus on specific segments of the feature space during the generation of output, making it sensitive to relevant features.

The experimentation process employed comprehensive schedule of redefining our model with different middle optimization layers to guarantee that the model is optimized for flexibility. We added the following features optimization layers to those image classification models that were used in our previous experimentation.

- Batch Normalization
- Dropout
- Attention
- Global Max Pooling

5.4 Implementation of DINAP Model

This paper uses a structured and methodological approach to experimentation. We fully observe the results of each iteration of our experiments while keeping in mind the goal of proposing a model that is highly efficient accurate and also less resource intensive. This paper then proposes a model, DINAP for Detection of Novel Attacks in Phishing. DINAP uses the triplet sampling approach where it takes 3 input images, an Anchor, a Positive, and a Negative. It uses VGG19 as a base image classifier with the classification layers removed in order to use it as an image feature extractor. It uses an Attention mechanism and Global Max Pooling as the middle layers for the function of feature optimization and uses various pre-processing techniques that allows it to get ready for classification.

5.5 Conclusion

This chapter extensively covered the implementation details of our proposed model, DINAP. We discussed the various experimentation that was conducted during the designing and implementation phases, offering a detailed look into our proposed methodology. The insights from this chapter will help us in our discussion about results and comparisons, giving us a good understanding of how well our proposed model DINAP works.

Chapter 6 Experimental Results and Discussion

In this chapter we will discuss the results of all the experimentation carried out while designing and implementing the DINAP model. Moreover, we will also compare the results of our DINAP model with the base case model VisualPhishNet [1] on VisualPhish dataset and VisualPhish dataset 2.0. DINAP and VisualPhishNet need to be trained twice. In first training, we train the model on the dataset and in the second phase of training we fine tune the model on the same dataset.

6.1 Results of Base Case Implementation

Following are the results that we got when we implemented the base case model:

Table 6.1: Results of VisualPhishNet Model on VisualPhish Dataset

The accuracy of VisualPhishNet model on VisualPhish Dataset before fine tuning and after fine tuning is presented here.

Model	Accuracy after 1st Training	Accuracy after Fine Tuning (2nd Training)
VisualPhishNet	55	77

We repeated the experimentation for up to 5 times and above-mentioned accuracies are the highest recorded. Although the accuracy mentioned in the paper after fine-tuning the model is 81%.

6.2 Results by Replacing Image Classifier of Base Model

Following are the results that we got when we used different base image classifiers with fine tuning the model:

Table 6.2: Results by Replacing Image Classifier

The accuracies of various image classification models on VisualPhish Dataset without fine tuning that we got throughout our experimentation is presented here.

Model	Accuracy without fine tuning (1st Training)
Xception	12
EfficientNetV2S	16

InceptionV3	19
EfficientNetV2B3	20
InceptionResNetV2	22
ResNet152V2	28
NASNetLarge	29
EfficientnetB0	36
NASNetMobile	36
MobileNetV2	39
DenseNet201	45
VGG19	56

The takeaway from the above experimentation was that the VGG19 base image classifier had slightly more accuracy without fine-tuning than the base model.

6.3 Results by adding Features Optimization Layers

Following are the results we got after adding data optimization layers (Batch Normalization & Dropout & Attention Layer) to the different base image classifiers that were used in previous experimentation.

Table 6.3: Results by adding Features Optimization Layers

The accuracies after adding various features optimization layers to different image classification models without fine tuning that we got throughout our experimentation is presented here.

Model	**Accuracy without fine tuning (1st Training)**
Xception with Batch Normalization + Dropout + Attention	15

InceptionV3 with Batch Normalization + Dropout + Attention	17
EfficientNetV2B3 with Batch Normalization + Dropout + Attention	20
ResNet152V2 with Batch Normalization + Dropout + Attention	26
EfficientNetV2S with Batch Normalization + Dropout + Attention	26
NASNetMobile with Batch Normalization + Dropout + Attention	28
NASNetLarge with Batch Normalization + Dropout + Attention	29
InceptionResNetV2 with Batch Normalization + Dropout + Attention	30
EfficientnetB0 with Batch Normalization + Dropout + Attention Layers	32
DenseNet201 with Batch Normalization + Dropout + Attention	32
MobileNetV2 with Batch Normalization + Dropout + Attention	32
VGG19 with Batch Normalization + Dropout + Attention	52

The takeaway from above experimentation was that, we need to adjust and make additions or remove one or more features optimization layers that are affecting the performance of the model.

6.4 Results of DINAP

Following are the results that we got when we implemented our proposed DINAP Model:

Table 6.4: Results of DINAP Model

The accuracy of DINAP model before fine tuning and after fine tuning is presented here.

Model	Accuracy without fine tuning (1st Training)	Accuracy after Fine Tuning (2nd Training)
DINAP	64	84

We repeated the experimentation for up to 5 times and above-mentioned accuracies are the highest recorded.

6.5 Comparison with VisualPhishNet

Our proposed model DINAP has shown better performance and accuracy than base model VisualPhish-Net [1]. Plus our model is also less resource intensive and highly accurate.

Table 6.5: Comparison of DINAP and VisualPhishNet on VisualPhish Dataset

The comparsion of accuracies of DINAP and VisualPhishNet model before fine tuning and after fine tuning is presented here.

Model	Accuracy without fine tuning (1st Training)	Accuracy after Fine Tuning (2nd Training)
VisualPhishNet	54	77
DINAP	64	84

6.6 Results on VisualPhish Dataset 2.0

DINAP has also shown better performance and accuracy than base model VisualPhishNet [1] on VisualPhish Dataset 2.0.

Table 6.6: Results of DINAP and VisualPhishNet on VisualPhish Dataset 2.0

The comparsion of accuracies of DINAP and VisualPhishNet model on the augmented dataset VisualPhish 2.0 is presented here.

Model	Accuracy without fine tuning (1st Training)	Accuracy after Fine Tuning (2nd Training)
VisualPhishNet	49	69
DINAP	56	75

6.7 Conclusion

In this chapter we discussed the results we got throughout our experimentation, designing and implementing the DINAP model. In terms of accuracy and efficiency, our DINAP model exceeds the VisualPhishNet [1] model. DINAP not only produces superior results but also uses fewer resources, demonstrating its potential as an efficient and resource-saving solution in the field of novel phishing detection. These findings highlight DINAP's practical benefits and its promising role in enhancing cybersecurity efforts.

Chapter 7 Future Work

We presented DINAP, a novel phishing website detection model that uses screenshots of the websites diverging from conventional methods of using the URL to detect whether the website is legitimate or phishing. We used the deep learning algorithm "Convolutional Neural Network (CNN)" to achieve high accuracy on detecting novel phishing websites. Notably, DINAP's accuracy surpasses that of the Visual Phish Model [1] when tested on both the Visual Phish Dataset and Visual Phish 2.0 Dataset.

We hope that our research will help the cyber security professionals and researchers to develop systems capable of swiftly and accurately detecting novel phishing threats. In future work, our focus will be on refining DINAP to minimize misclassifications, thereby enhancing its effectiveness in real-world applications.

Bibliography

[1] S. Abdelnabi, K. Krombholz, and M. Fritz, "Visualphishnet: Zero-day phishing website detection by visual similarity," in *Proceedings of the 2020 ACM SIGSAC conference on computer and communications security*, pp. 1681–1698, 2020.

[2] A. Aljofey, Q. Jiang, Q. Qu, M. Huang, and J.-P. Niyigena, "An effective phishing detection model based on character level convolutional neural network from url," *Electronics*, vol. 9, no. 9, p. 1514, 2020.

[3] M. A. Adebowale, K. T. Lwin, and M. A. Hossain, "Intelligent phishing detection scheme using deep learning algorithms," *Journal of Enterprise Information Management*, vol. 36, no. 3, pp. 747–766, 2023.

[4] V. Shahrivari, M. M. Darabi, and M. Izadi, "Phishing detection using machine learning techniques," *arXiv preprint arXiv:2009.11116*, 2020.

[5] M. Korkmaz, E. Koçyiğit, Ö. Şahingöz, and B. Diri, "A hybrid phishing detection system by using deep learning-based url and content analysis," *Elektronika ir Elektrotechnika*, vol. 28, no. 5, 2022.

[6] S. Y. Yerima and M. K. Alzaylaee, "High accuracy phishing detection based on convolutional neural networks," in *2020 3rd International Conference on Computer Applications & Information Security (ICCAIS)*, pp. 1–6, IEEE, 2020.

[7] A. Mughaid, S. AlZu'bi, A. Hnaif, S. Taamneh, A. Alnajjar, and E. A. Elsoud, "An intelligent cyber security phishing detection system using deep learning techniques," *Cluster Computing*, vol. 25, no. 6, pp. 3819–3828, 2022.

[8] E. Zhu, Z. Chen, J. Cui, and H. Zhong, "Moe/rf: A novel phishing detection model based on revised multiobjective evolution optimization algorithm and random forest," *IEEE Transactions on Network and Service Management*, vol. 19, no. 4, pp. 4461–4478, 2022.

[9] A. El Aassal, S. Baki, A. Das, and R. M. Verma, "An in-depth benchmarking and evaluation of phishing detection research for security needs," *IEEE Access*, vol. 8, pp. 22170–22192, 2020.

[10] M. Al-Sarem, F. Saeed, Z. G. Al-Mekhlafi, B. A. Mohammed, T. Al-Hadhrami, M. T. Alshammari, A. Alreshidi, and T. S. Alshammari, "An optimized stacking ensemble model for phishing websites detection," *Electronics*, vol. 10, no. 11, p. 1285, 2021.

[11] D.-J. Liu, G.-G. Geng, X.-B. Jin, and W. Wang, "An efficient multistage phishing website detection model based on the case feature framework: Aiming at the real web environment," *Computers & Security*, vol. 110, p. 102421, 2021.

[12] C. C. L. Tan, K. L. Chiew, K. S. Yong, Y. Sebastian, J. C. M. Than, and W. K. Tiong, "Hybrid phishing detection using joint visual and textual identity," *Expert systems with applications*, vol. 220, p. 119723, 2023.

[13] J. Rashid, T. Mahmood, M. W. Nisar, and T. Nazir, "Phishing detection using machine learning technique," in *2020 first international conference of smart systems and emerging technologies (SMARTTECH)*, pp. 43–46, IEEE, 2020.

[14] R. Paturi, L. Swathi, K. S. Pavithra, R. Mounika, and C. Alekhya, "Detection of phishing attacks using visual similarity model," in *2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, pp. 1355–1361, IEEE, 2022.

[15] L. Lakshmi, M. P. Reddy, C. Santhaiah, and U. J. Reddy, "Smart phishing detection in web pages using supervised deep learning classification and optimization technique adam," *Wireless Personal Communications*, vol. 118, no. 4, pp. 3549–3564, 2021.

[16] S. Al-Ahmadi, A. Alotaibi, and O. Alsaleh, "Pdgan: Phishing detection with generative adversarial networks," *IEEE Access*, vol. 10, pp. 42459–42468, 2022.

[17] M. Sánchez-Paniagua, E. Fidalgo, E. Alegre, and R. Alaiz-Rodríguez, "Phishing websites detection using a novel multipurpose dataset and web technologies features," *Expert Systems with Applications*, vol. 207, p. 118010, 2022.

[18] J. Zhou, H. Cui, X. Li, W. Yang, and X. Wu, "A novel phishing website detection model based on lightgbm and domain name features," *Symmetry*, vol. 15, no. 1, p. 180, 2023.

[19] M. S. Gayathri *et al.*, "Detection of phishing attack using gan with rfc," *Rivista Italiana di Filosofia Analitica Junior*, vol. 14, no. 2, pp. 479–487, 2023.

[20] A. Butnaru, A. Mylonas, and N. Pitropakis, "Towards lightweight url-based phishing detection," *Future internet*, vol. 13, no. 6, p. 154, 2021.